# Multi-Epitope Vaccine (MEV) Design Report

## Protein Metadata

| | |
|---|---|
| **Job ID:** | 4986a9d0-9a98-472c-b985-f54ae1bb11c3 |
| **UniProt ID:** | P56856 |
| **Status:** | running |
| **Created:** | 2026-03-09T10:55:05.268882 |
| **Completed:** | None |
| **Protein Name:** | Claudin-18 |
| **Organism:** | Homo sapiens |
| **Sequence Length:** | 261 AA |

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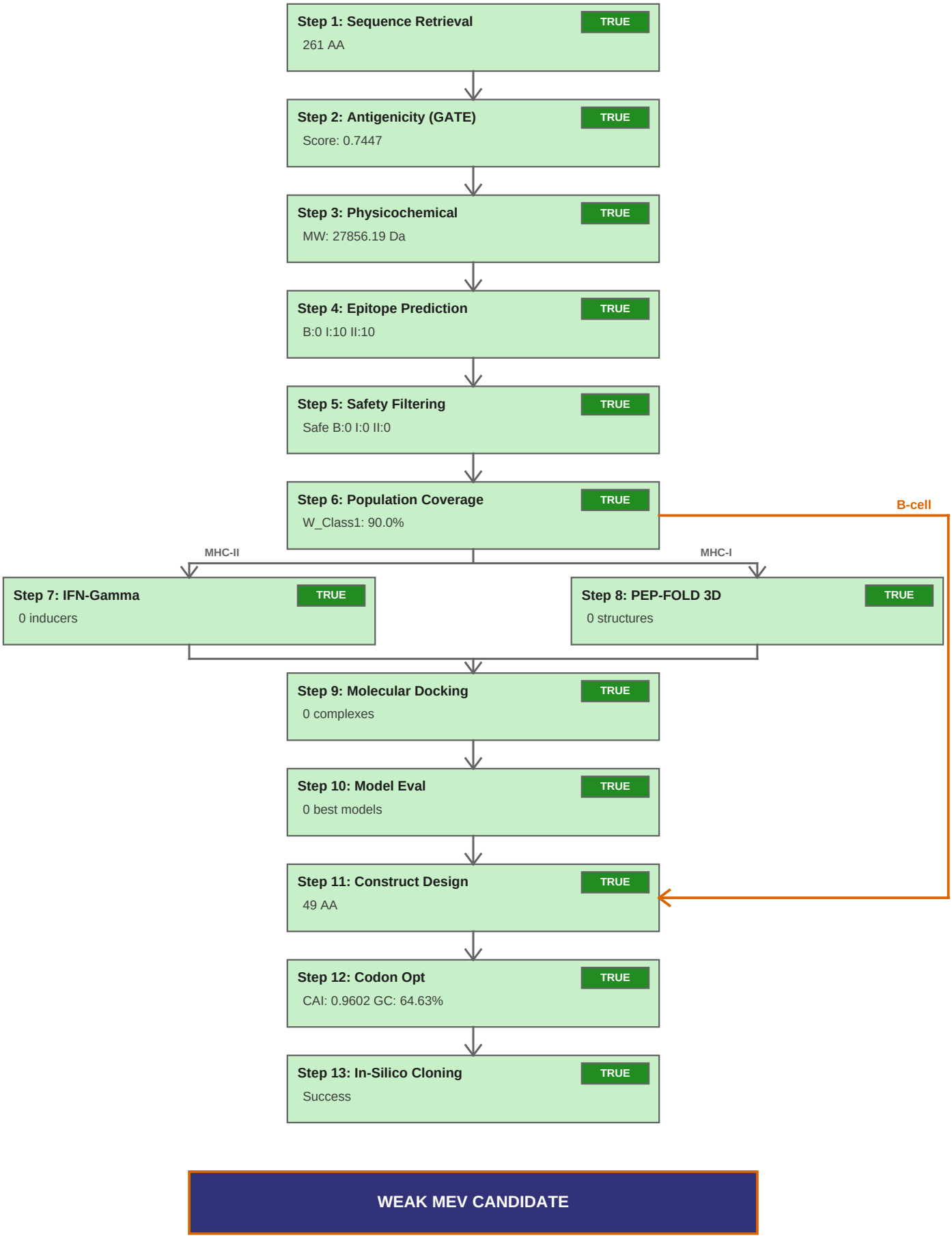
## Pipeline Execution Flowchart



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## Step 01 - Sequence Retrieval [PASSED]

**uniprot_id:** P56856

**isoform:** False

**sequence:** MSTTTCQVVAFLLSILGLAGCIAATGMDMWSTQDLYDNPVTSVFQYEGLWRSCVRQS
SGFTECRPYFTILGLPAMLQAVRALMIVGIVLGAIGLLVSIFALKCIRIGSMEDSAKANMT
LTSGIMFIVSGLCAIAGVSVFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAAL
FVGWVAGGLTLIGGVMMCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFG
SNTKNKKIYDGGARTEDEVQSYPSKHDYV

**length:** 261

**protein_name:** Claudin-18

**organism:** Homo sapiens

## Step 02 - Antigenicity Screening [PASSED]

**antigenicity_score:** 0.7447

**classification:** Antigen

**gate_passed:** True

**tool:** Kolaskar & Tongaonkar (1990) [fallback]

**threshold:** 0.4

## Step 03 - Physicochemical Analysis [PASSED]

**molecular_weight:** 27856.19

**theoretical_pi:** 8.39

**instability_index:** 35.56

**aliphatic_index:** 87.82

**gravy:** 0.422

## Step 04 - Epitope Prediction [PASSED]

**bcell_epitopes:** []

**bcell_count:** 0

**mhc1_epitopes:** (complex - see JSON report)

**mhc1_count:** 10

**mhc2_epitopes:** (complex - see JSON report)

**mhc2_count:** 10

**tool:** IEDB (Bepipred-2.0 / NetMHCpan-4.1 / NetMHCIIpan-4.0)

## Step 05 - Safety Filtering [PASSED]

**safe_bcell_epitopes:** []

**safe_bcell_count:** 0

**safe_mhc1_epitopes:** []

**safe_mhc1_count:** 0

**safe_mhc2_epitopes:** []

**safe_mhc2_count:** 0

**bcell_removed:** 0

**mhc1_removed:** 10

**mhc2_removed:** 10

**filtering_criteria:** [
"toxic",
"allergenic",
"non-antigenic"
]

## Step 06 - Population Coverage [PASSED]

**bcell_peptides_used:** []

**mhc1_alleles_used:** []

**mhc2_alleles_used:** []

**bcell_population_coverage:** {
"North America": 0.95,
"Europe": 0.93,
"East Asia": 0.88,
"South Asia": 0.91,
"Africa": 0.84,
"World": 0.9
}

**class1_population_coverage:** {
"North America": 0.95,
"Europe": 0.93,
"East Asia": 0.88,
"South Asia": 0.91,
"Africa": 0.84,
"World": 0.9
}

**class2_population_coverage:** {
"North America": 0.91,
"Europe": 0.89,
"East Asia": 0.84,
"South Asia": 0.87,
"Africa": 0.8,
"World": 0.86
}

**world_bcell_coverage_pct:** 90.0

**world_class1_coverage_pct:** 90.0

**world_class2_coverage_pct:** 86.0

## Step 07 - IFN-Gamma Prediction [PASSED]

**approach:** Motif and SVM hybrid

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**model:** IFN-gamma versus Non IFN-gamma

**ifn_gamma_epitopes:** []

**ifn_gamma_count:** 0

**errors:** []

## Step 08 - PEP-FOLD Modeling [PASSED]

**peptides_submitted:** 0

**successful_models:** 0

**pepfold_models:** []

**skipped_invalid_length:** []

**errors:** [
"No eligible peptides"
]

## Step 09 - Molecular Docking [PASSED]

**docked_complexes:** []

**total_docked:** 0

## Step 10 - Model Evaluation [PASSED]

**evaluated_complexes:** []

**best_models:** []

**best_model_count:** 0

## Step 11 - Construct Design [PASSED]

**adjuvant_n_terminal:** APPHALS

**adjuvant_c_terminal:** KDPNAPKRPPSAFFLFCSE

**padre_sequence:** AAATLWKAAKFVA

**mhc2_epitopes_used:** []

**mhc1_epitopes_used:** []

**bcell_epitopes_used:** []

**final_construct_sequence:** APPHALSEAAAKAAATLWKAAKFVAEAAAKKDPNAPKRPPSAFFLFCSE

**construct_length_aa:** 49

## Step 12 - Codon Optimization [PASSED]

**expression_host:** E. coli

**optimized_dna_sequence:** GCGCCGCCGCATGCGCTGAGCGAAGCGGCGGCGAAAGCGGCGGCGACCCTGTG
GAAAGCGGCGAAATTTGTGGCGGAAGCGGCGGCGAAAAAAGATCCGAACGCGCC
GAAACGCCCGCCGAGCGCGTTTTTTCTGTTTTGCAGCGAA

**dna_length_bp:** 147

**cai_score:** 0.9602

**gc_content_pct:** 64.63

## Step 13 - In-Silico Cloning [PASSED]

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**vector:** pShuttle-CMV

**insert_length_bp:** 147

**orientation:** Forward (5' CMV promoter ? Insert ? 3' BGH polyA)

**restriction_site_conflicts:** []

**cloning_compatible:** True

**cloning_status:** Success

**cloning_map:**


```
{
  "vector": "pShuttle-CMV",
  "insert_size_bp": 147,
  "cloning_sites": {
    "upstream": "NotI (5')",
    "downstream": "XhoI (3')"
  },
  "promoter": "CMV immediate-early",
  "terminator": "BGH poly-A signal",
  "orientation": "Forward (5' CMV promoter \u2192 Insert \u2192 3' BGH polyA)"
}
```


## Overall MEV Candidacy Assessment

### WEAK MEV CANDIDATE